

TIH-IOT Chanakya 2023 - Fellowship Report

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The objective of this fellowship was to explore and enhance the field of emotion recognition in text data using various NLP and large language model (LLM) concepts and models which would be further used for the chatbot application. I was responsible for several key tasks. First, I labeled a collected dataset with basic human emotions, including joy, sadness, anger, fear, shame, guilt, love, and surprise. This labeling process involved carefully analyzing each text entry to accurately categorize the expressed emotions. Next, I ran the Long Short-Term Memory (LSTM) model that was created using this labeled dataset. This step included training and testing the model to ensure its accuracy and reliability in detecting and classifying emotions in text. The LSTM allowed for effective feature extraction and sequence learning, which are crucial for understanding the nuanced expressions of emotions in textual data.

Pre-processing work :

At first, I performed essential pre-processing work, which involved sorting the comments from A to Z, using filters to remove links in the comments, removing unwanted blank spaces, and eliminating URLs and links. These pre-processing steps ensured the dataset's cleanliness and consistency, which is crucial for accurate model training.

- Sorting the comments from A to Z
- Using filter to remove the links in the comments
- Removing unwanted blank spaces
- Removing the urls,links

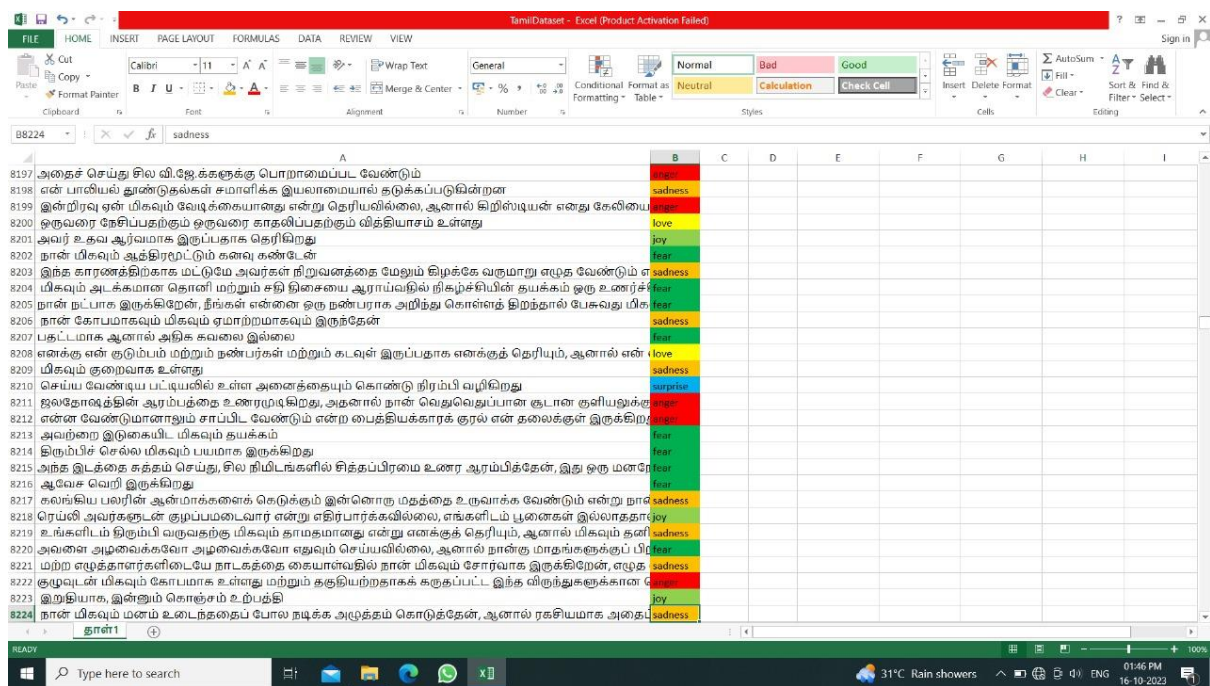
THE DATASET AFTER PREPROCESSING:

	A	B
25880	Past experience is where I spend most of my week under stress.	
25881	I feel that pain now and I fear that I will have to endure it for many more years	
25882	I feel certain that you and I have something of interest	
25883	usually like to stay at home and started worrying about this	
25884	I need the feeling that someone is committed to me, wants to keep me strong, and do things with me.	
25885	As I write this I realize, Oh, don't worry Leanne, your prince will be beautiful one day.	
25886	Didn't feel comfortable	
25887	told my parents that I no longer had romantic feelings or desires for affection with my boyfriend and that I wanted to break up with him	
25888	Feeling unloved and knowing being hated	
25889	tried to pinpoint the exact thought that made me feel stupid after presenting a task	
25890	Many times I feel so generous that I keep track of what I have given them or given up for them, and calculate what they owe me.	
25891	PARTY BABY DON'T CLOSE YOUR EYES WATCH ME HOTSTABODY EXCODICAPEDY BETWEEN THE TOMORROW IS DELICIOUS AND YOU WANT	
25892	I still feel damaged	
25893	Kim and I tried hard to avoid his insults	

LABELLING PROCESS:

The labeling process was a critical component of the project on emotion detection in text. This task involved meticulously analyzing a collected dataset to categorize each text entry according to basic human emotions, such as joy, sadness, anger, fear, shame, guilt, love, and surprise. Initially, I reviewed each comment to understand the context and the emotion conveyed. This required a keen understanding of language nuances and emotional expressions. Once the emotion was identified, I assigned the appropriate label to the text entry. To ensure consistency and accuracy, I followed a standardized set of guidelines and criteria for each emotion. This careful and systematic approach to labeling was essential for creating a high-quality dataset, which is crucial for training the Long Short-Term Memory (LSTM) model used in the project. The labeled dataset enabled the model to learn and accurately detect emotions in textual data.

THE DATASET AFTER PREPROCESSING:



	A	B	C	D	E	F	G	H	I
8197	அதைச் செய்து சில விடேகைகளுக்கு பொறாமைப்பட வேண்டும்	anger							
8198	என் பானியல் தூண்டுதல்கள் சமாளிக்க இயலாமையால் தடுக்கப்படுகின்றன	sadness							
8199	இன்றிரவு ஏன் மிகவும் வேடிக்கையானது என்று தெரியவில்லை, ஆனால் கிறிஸ்டியன் என்ன கேலியை	anger							
8200	ஒருவரை நேசிப்பதற்கும் ஒருவரை காதலிப்பதற்கும் வித்தியாசம் உள்ளது	love							
8201	அவர் உதவ ஆர்வமாக இருப்பதாக தெரிகிறது	joy							
8202	நான் மிகவும் ஆத்திரமூட்டும் கனவு கண்டேன்	fear							
8203	இந்த காரணத்திற்காக மட்டுமே அவர்கள் நிறுவனத்தை மேலும் கிழக்கே வருமாறு எழுத வேண்டும் எ	sadness							
8204	மிகவும் அடக்கமான தொனி மற்றும் சதி நிசையை ஆராய்வதில் நிகழ்ச்சியின் தயக்கம் ஒரு உணர்ச்சி	fear							
8205	நான் நட்பாக இருக்கிறேன், நீங்கள் என்னை ஒரு நண்பராக அறிந்து கொள்ளத் திறந்தால் பேசுவது மிக	fear							
8206	நான் கோபமாகவும் மிகவும் ஏமாற்றமாகவும் இருந்தேன்	sadness							
8207	பதட்டமாக ஆனால் அதிக கவலை இல்லை	fear							
8208	எனக்கு என் குடும்பம் மற்றும் நண்பர்கள் மற்றும் கடவுள் இருப்பதாக எனக்குத் தெரியும், ஆனால் என்	love							
8209	மிகவும் குறைவாக உள்ளது	sadness							
8210	செய்ய வேண்டிய பட்டியலில் உள்ள அனைத்தையும் கொண்டு நிரம்பி வழிவது	surprise							
8211	ஜல்தோஷத்தின் ஆரம்பத்தை உணரமுடிவது, அதனால் நான் வெதுவெதுப்பான சூடான குளியலுக்கு	anger							
8212	என்ன வேண்டுமானாலும் சாப்பிட வேண்டும் என்ற பைத்தியக்காரக் குரல் என் தலைக்குள் இருக்கிறத	anger							
8213	அவற்றை இடுகையில் மிகவும் தயக்கம்	fear							
8214	கிறும்பிச் செல்ல மிகவும் பயமாக இருக்கிறது	fear							
8215	அந்த இடத்தை சுத்தம் செய்து, சில நிமிடங்களில் சித்தப்பிரமை உணர் ஆரம்பித்தேன், இது ஒரு மனநோ	fear							
8216	ஆவேச வெறி இருக்கிறது	fear							
8217	கலங்கிய பலரின் ஆன்மாக்களைக் கெடுக்கும் இன்னொரு மதத்தை உருவாக்க வேண்டும் என்று நாடி	sadness							
8218	ரெய்லி அவர்களுடன் குழப்பமடைவார் என்று எதிர்பார்க்கவில்லை, எங்களிடம் பூனைகள் இல்லாததா	joy							
8219	உங்களிடம் கிறும்பி வருவதற்கு மிகவும் தாமதமானது என்று எனக்குத் தெரியும், ஆனால் மிகவும் தனி	sadness							
8220	அவரை அழைக்கவோ அழைக்கவோ எதுவும் செய்யவில்லை, ஆனால் நான் மாதங்களுக்குப் பி	fear							
8221	மற்ற எழுத்தாளர்களிடையே நாடகத்தை கையாள்வதில் நான் மிகவும் சோர்வாக இருக்கிறேன், எழுத	sadness							
8222	சூழ்வுடன் மிகவும் கோபமாக உள்ளது மற்றும் தகுதியற்றதாகக் கருதப்பட்ட இந்த விருந்துகளுக்கான	anger							
8223	இறுதியாக, இன்னும் கொஞ்சம் உற்பத்தி	joy							
8224	நான் மிகவும் மனம் உடைந்ததைப் போல நடிக்க அழுத்தம் கொடுத்தேன், ஆனால் ரகசியமாக அதை	sadness							

EMOTION DATASET DESCRIPTION:

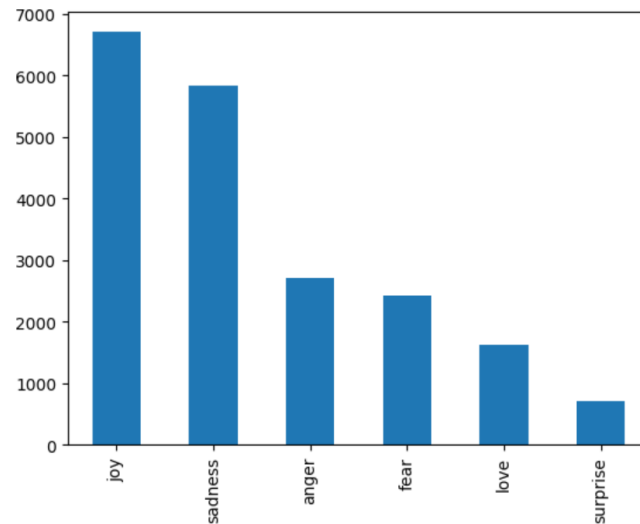
The dataset collected as a total of approx 20000 sentences with emotion distribution as following:

Table of Sentences Based on Emotions:

col_0	Count	Total	Percentage
Emotions			
anger	2698	2698	13.492024
fear	2421	2421	12.106816
joy	6702	6702	33.515027
love	1630	1630	8.151223
sadness	5831	5831	29.159374
surprise	715	715	3.575536
Total	19997	19997	100.000000

Emotion Distribution in the Dataset:

The bar chart above illustrates the distribution of emotions within the labeled dataset. The dataset shows a significant prevalence of the emotion "joy," with over 6,000 occurrences, followed by "sadness" with more than 5,000 entries. Other emotions like "anger" and "fear" have moderate representation, while "love" and "surprise" are less frequent. This distribution highlights the varying frequencies of different emotional expressions in the collected textual data, which can impact the training and performance of the LSTM model in detecting and classifying these emotions accurately.



LSTM Model

The LSTM model used in the project is a hybrid deep learning architecture designed to effectively detect and classify emotions in text. This model leverages the strengths of Long Short-Term Memory (LSTM) networks to capture the nuanced features and sequential dependencies in textual data.

```
model = Sequential()
model.add(Embedding(input_dim=len(tokenizer.word_index) + 1,
                    output_dim=128, input_length=max_length))
model.add(LSTM(256+1))
model.add(Flatten())
model.add(Dense(units=128, activation="relu"))
model.add(Dense(units=len(one_hot_labels[0]), activation="softmax"))

model.compile(optimizer="adam", loss="categorical_crossentropy", metrics=["accuracy"])
history=model.fit(xtrain, ytrain, epochs=15, batch_size=32, validation_data=(xtest, ytest))
```

TRAINING:

```
Epoch 1/15
400/400 [=====] - 22s 49ms/step - loss: 1.2150 - accuracy: 0.5394 - val_loss: 0.6589 - val_accuracy: 0.7728
Epoch 2/15
400/400 [=====] - 7s 18ms/step - loss: 0.3757 - accuracy: 0.8731 - val_loss: 0.4053 - val_accuracy: 0.8637
Epoch 3/15
400/400 [=====] - 4s 10ms/step - loss: 0.1669 - accuracy: 0.9441 - val_loss: 0.3221 - val_accuracy: 0.8919
Epoch 4/15
400/400 [=====] - 5s 12ms/step - loss: 0.1027 - accuracy: 0.9652 - val_loss: 0.3370 - val_accuracy: 0.8925
Epoch 5/15
400/400 [=====] - 5s 12ms/step - loss: 0.0663 - accuracy: 0.9773 - val_loss: 0.2957 - val_accuracy: 0.9031
Epoch 6/15
400/400 [=====] - 4s 9ms/step - loss: 0.0477 - accuracy: 0.9843 - val_loss: 0.4287 - val_accuracy: 0.8922
Epoch 7/15
400/400 [=====] - 4s 10ms/step - loss: 0.0405 - accuracy: 0.9855 - val_loss: 0.3761 - val_accuracy: 0.8978
Epoch 8/15
400/400 [=====] - 5s 12ms/step - loss: 0.0274 - accuracy: 0.9902 - val_loss: 0.4037 - val_accuracy: 0.8988
Epoch 9/15
400/400 [=====] - 4s 10ms/step - loss: 0.0287 - accuracy: 0.9905 - val_loss: 0.4077 - val_accuracy: 0.9056
Epoch 10/15
400/400 [=====] - 4s 10ms/step - loss: 0.0316 - accuracy: 0.9894 - val_loss: 0.3976 - val_accuracy: 0.8991
Epoch 11/15
400/400 [=====] - 4s 10ms/step - loss: 0.0234 - accuracy: 0.9924 - val_loss: 0.4249 - val_accuracy: 0.8953
Epoch 12/15
400/400 [=====] - 4s 11ms/step - loss: 0.0214 - accuracy: 0.9923 - val_loss: 0.5064 - val_accuracy: 0.8934
Epoch 13/15
400/400 [=====] - 4s 9ms/step - loss: 0.0216 - accuracy: 0.9930 - val_loss: 0.4176 - val_accuracy: 0.9062
Epoch 14/15
400/400 [=====] - 4s 9ms/step - loss: 0.0221 - accuracy: 0.9922 - val_loss: 0.4043 - val_accuracy: 0.8994
Epoch 15/15
400/400 [=====] - 4s 11ms/step - loss: 0.0160 - accuracy: 0.9940 - val_loss: 0.4901 - val_accuracy: 0.8963
```

TESTING:

```
 model.evaluate(xtest, ytest)

100/100 [=====] - 1s 5ms/step

loss: 0.0477 - accuracy: 0.9843
```

PREDICTION:

The given input text, "She is excited about her new job," is preprocessed and transformed into a sequence suitable for the model. The model then predicts the emotion, and the result, "joy," indicates that the model has successfully identified the emotion conveyed in the text.

```
[17] input_text = "She is excited about her new job"

# Preprocess the input text
input_sequence = tokenizer.texts_to_sequences([input_text])
padded_input_sequence = pad_sequences(input_sequence, maxlen=max_length)
prediction = model.predict(padded_input_sequence)
predicted_label = label_encoder.inverse_transform([np.argmax(prediction[0])])
print(predicted_label)

1/1 [=====] - 0s 20ms/step
['joy']
```

CONCLUSION:

Overall, the project on emotion detection in text involved multiple critical steps, including the careful labeling of a diverse dataset with basic human emotions and the implementation of a LSTM model for accurate emotion classification. The labeling process ensured a high-quality dataset, while the pre-processing steps like sorting comments, removing links, and eliminating unwanted spaces prepared the data for effective analysis. The LSTM model leveraged the strengths of LSTMs to capture and interpret emotional nuances in textual data.

I am deeply grateful to my mentor, Dr. Radha Senthil Kumar, Research Scholar Nasrin Banu, and TiH-IOT - IIT Bombay for the invaluable opportunity to delve into the field of natural language processing and machine learning. Their guidance and support have been instrumental in expanding my knowledge and understanding of these advanced technologies. This experience has significantly enriched my skills and insights, and I am truly appreciative of the chance to learn and grow under their mentorship.